

Cause and Effect:

Texts That Explain Reasons and Results

Understand What Cause and Effect Are

A **cause** is what makes an event, or effect, happen. An **effect** is what happens as a result of an event, or cause.

Four Basics of Good Cause and Effect

- 1 The main point reflects the writer's purpose: to explain causes, effects, or both.
- 2 If the purpose is to explain causes, the writing presents real causes; if the purpose is to explain effects, the writing presents real effects.
- 3 It provides details and examples that support and explain the causes and/or effects.
- 4 It can be organized in different ways—order of importance, space order, or time order—depending on its purpose.

In the following paragraph, the numbers and colors correspond to the Four Basics of Good Cause and Effect.

Followers of college football recognize several outstanding programs, including Notre Dame, Michigan, Ohio State, and Florida State. However, in the past decade, one school in particular has dominated the college football rankings, with championships in 2009, 2011, 2012, and 2015: The University of Alabama, known as the Crimson Tide. 1 What has led to Alabama's dominance? 2 First, there is a tradition of winning at Alabama, 3 which has had sixteen national titles in its history, six under the legendary Paul "Bear" Bryant in the 1960s and 1970s, and four under current coach Nick Saban. 2 That tradition of winning has led to

4 Points arranged in order of importance.

4 Points arranged in order of importance.

outstanding recruiting: Alabama has brought some of the best talent in the country to the university. **3** The ability to recruit well means that Alabama's teams have depth: they can substitute outstanding players throughout the game, bringing fresh energy to the field. **2** Another reason for Alabama's success is the financial investment they have made in the team. **3** According to the NCAA, Alabama spent just under \$1 million in 2012–2013 on recruiting, and in 2016, coach Nick Saban was the highest paid college football coach in the country, earning a staggering \$7.09 million. Coach Saban does not coach alone: the university employs fourteen assistant coaches as well. **2** But perhaps more important than the tradition and the money spent, Alabama has a winning philosophy embraced by all the coaches: develop a solid defense, and treat every opponent with respect. **3** In a recent broadcast on ESPN, former Alabama quarterback Greg McElroy noted that Coach Saban and his staff expect the same preparation each week, both for the lowest-ranked opponent and for the championship game. **1** With tradition, financial support, strong coaching, and a solid team philosophy, Alabama has become a championship team.

Teaching Tip: Use the chapter title and references to effect on page 245 to discuss the difference between *effect* (noun) and *affect* (verb).

Teaching Tip: You might point out that mistaken causes or effects are examples of logical fallacies. Avoiding logical fallacies is an important part of critical thinking. An excellent resource to help students understand logical fallacies is www.fallacyfiles.org.

Understanding cause and effect helps you read and write in many situations. In college, you may need to read a book that analyzes the causes of the Second World War, or you might need to describe the effects of malnutrition for a health class. If you work for city government, you might read a report discussing the results of new traffic patterns near public schools, or you might be asked to investigate the causes of an increase in crime in one part of town. In everyday life, you may read about the possible side effects of an antibiotic you are taking, or you might explain to a child the consequences of bad behavior. All of these activities require an understanding of cause and effect.

Cause and Effect in the Classroom: Using Diagrams to Show a Chain of Events

In Chapter 1, we met Jackie. In Jackie's Student Development class, the instructor asked the students to consider the *chain of events* that led to a failure in their lives. Jackie decided to write about her failure in her first college English class. She had learned the Four Basics of Good Cause and Effect analysis, so she brainstormed the following list of causes for her failure.

CAUSES OF MY F GRADE:

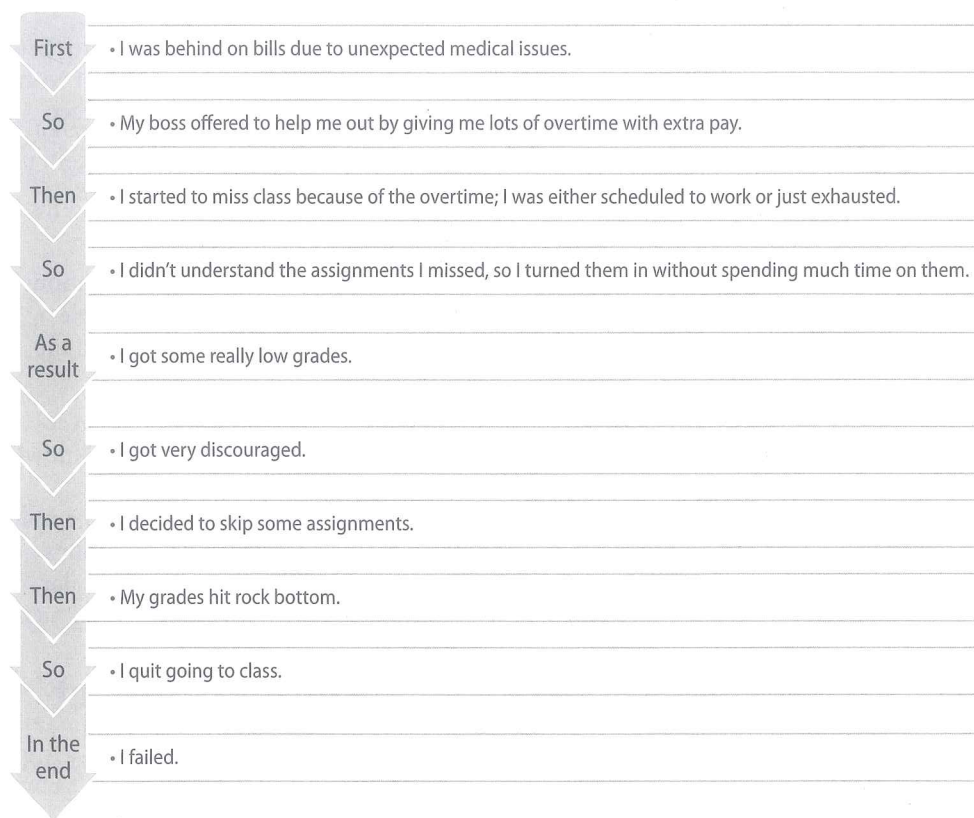
missed classes

not enough time spent on assignments (no editing or proofreading)

late assignments

didn't understand assignments

Jackie wanted to show that she had made mistakes in her class, and that was why she failed. She knew all of her reasons were real causes, but she did not know how to show a chain of events. She took the assignment to her English teacher, who explained that a “chain of events” is a series of cause-and-effect relationships. The effect of one initial event causes the next event to happen, which in turn causes the next step in the chain. After talking with her teacher, Jackie realized that she could make a graphic organizer to help her show the chain of events.

**Teaching Tip:**

This chart is one example of a graphic organizer for cause-and-effect relationships. Many students find graphic organizers helpful. Additional examples of graphic organizers can be found at www.teachnology.com/worksheets/graphic/.

The diagram helped Jackie understand the “chain of events”: each effect became a cause that led to another effect. The causes weren’t just separate events; each problem could be traced back to her original decision to work overtime in order to pay her medical bills.

PRACTICE 12-1 Thinking about Chains of Events

Using Jackie’s diagram as a model, consider what chain of events might lead a struggling student to become more successful.

Reading Cause and Effect

When you are reading, look for clues that a writer has developed the topic using cause and effect, such as the transition words in the box on page 254. Also look for key words such as “why,” “results,” “effects,” “source,” or “cause” in the title, subtitles, or topic sentences. Once you have identified a focus on causes or effects, you can read critically by asking these questions:

- *What is the writer’s purpose?* Is it to show causes, effects, or both?
- *Are the causes and effects logical and supported by evidence?* Make sure that the writer has not confused something that happened before an event with a real cause or something that happened after an event with a real effect. For example, if someone eats pizza on Monday and gets the flu on Tuesday, eating the pizza is not the cause of the flu just because it happened first, nor is the flu the effect of eating pizza. The relationship between pizza and flu is not logical or supported by evidence.
- *Has the writer oversimplified the causes of a complex issue?* Many situations and problems are complex, and we cannot reduce the causes to a single factor. As you saw in Jackie’s situation earlier, failure in a class is often the result of many causes, not a single event.
- *Has the writer acknowledged all possible results?* Just as a writer may oversimplify the causes of a particular problem, a writer may refuse to acknowledge all possible results of a decision or course of action. For example, a blogger who chooses to focus only on the negative effects of chemotherapy for cancer patients—including nausea, hair loss, and potential infections—will give an incomplete picture of the risks and rewards of this particular medical treatment.

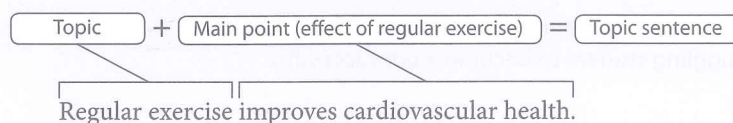
First Basic: Main Point in Cause and Effect

The **main point** introduces causes, effects, or both and the reason for writing about them. To help you discover the main point, complete the following sentences:

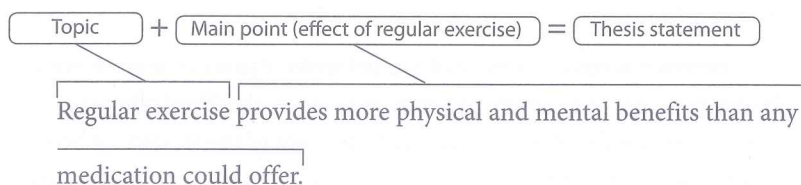
(The topic) causes (or caused) . . .

(The topic) results in (or resulted in) . . .

Here is an example of a topic sentence for a paragraph:



Remember that the main point for an essay can be a little broader than one for a paragraph.



Whereas the topic sentence focuses on just one major benefit of regular exercise, the thesis statement considers multiple benefits.

PRACTICE 12-2 Identify the Main Point in Cause and Effect

1. The topic sentence in the sample paragraph on pages 245–46 is highlighted in blue. Identify the topic, the main idea, and the focus (cause, effect, or both).
2. Find the main point in the sample paragraph on page 256. Identify the topic, the main idea, and the focus (cause, effect, or both).
3. Find the main point in the essay by Tyler Dashner on pages 260–64. Identify the topic, the main idea, and the focus (cause, effect, or both).

Second Basic: Real Causes and Effects

Major support in a cause-and-effect essay includes a list of causes and/or effects. It is important to think carefully when you are determining causes and effects; events, conditions, and problems often have multiple causes or effects, as the following chart illustrates.

One cause → one effect	Thunderstorm → game was canceled
One cause → multiple effects	Hurricane → flooding, loss of electricity, loss of homes and buildings, higher insurance, etc.
Multiple causes → one effect	Lack of exercise, consumption of junk food and sweets, stress, sleeplessness → weight gain
Multiple causes → multiple effects	Injuries, academic probation, changes in coaching staff, heat wave → 0–6 record, NCAA investigation, coach fired
Cause → effect → effect → effect → effect	Lost cell phone → missed a key business call → didn't get a contract → didn't get the promotion

PRACTICE 12-3**Recognize Support and Details**

The topic sentence has been double underlined in the following paragraph. Study the paragraph and underline and number (1, 2, 3, etc.) the major supporting details.

- Between weeks 11 and 12 of the semester, a strange phenomenon happens across our community college campus: the “third quarter drop out.” During these weeks, as many as 10 percent of the students who had been regularly attending disappear. What happens? There are three primary causes for the attendance slump. One powerful reason for a student to stop coming to class is discouragement. The midterms are finished, and if a student’s grade is not good at this point, there may be little chance to improve the overall grade in the course. If the deadline to drop the course has already passed, the student may see no reason to continue without a chance for success. In addition, pressures of life, family, job, and finances may become overwhelming to a student. Some students may stop attending class to pick up extra overtime pay at work, while others become stressed due to illness or lack of transportation to and from class. Finally, and perhaps most importantly, many students do not know how to get the help they need to finish the class, even though resources are available. For example, many colleges provide carpool services for students with transportation issues, while others will work with students to handle financial issues through payment plans. Many colleges make tutoring and writing centers available to students without charge, but students may not know how to find these services on their own. While no one program or service will stop the attendance drop, instructors and college personnel can work with students to identify issues early and direct them to the help that they need.

Avoid Logical Fallacies: Post Hoc and Slippery Slope

As you choose causes or effects, make sure you avoid two common **logical fallacies**, or mistakes in reasoning.

The **post hoc fallacy** comes from a Latin phrase, *post hoc ergo propter hoc*, that literally means “after this, therefore because of this.” We make this mistake when we look at two events and assume that the one that occurred first caused the second one. Superstitions, for example, illustrate the post hoc fallacy. Imagine a student whose teacher asks why she did not pass a test. The student responds, “Well, a black cat crossed in front of me on the way to school—and I just had bad luck.” While the student may have seen a black cat crossing on the way to school, there is no logical reason to believe that the cat caused the student’s problems on the test.

PRACTICE 12-4 Identify Post Hoc Mistakes

Evaluate each of the statements of cause below. Are any of them examples of the post hoc fallacy? Explain.

1. John spent the past two weeks partying and getting in shape for football try-outs; he didn't even open his books. That's why he failed the test today.
2. John forgot to wear his lucky shirt to class. That's why he failed the test.
3. The professor forgot to return John's homework last week. That's why he failed the test today.
4. John injured his knees during a workout a few weeks ago. That's why he failed the test.

The **slippery slope fallacy** occurs when someone suggests certain effects will occur following an action or event, but there is no logical basis for that belief. For example, a student who receives an F grade on a daily homework assignment might plead with her instructor this way: "If I make an F on this assignment, I will never be able to transfer to the university, graduate, or get a good job." Because homework counts for only 5 percent of the grade for this class, the student has presented an illogical effect; in other words, she is going down a slippery slope. A single homework assignment cannot lead to all the negative outcomes she suggested. *Post hoc thinking* leads to illogical cause statements; *slippery slope thinking* leads to illogical effect statements.

PRACTICE 12-5 Recognize Slippery Slope Statements

Study the following statements. Determine if the writer has committed a slippery slope fallacy in any of them.

1. I cannot believe I just ate that slice of pizza. There is no way I will be able to fit into my prom dress.
2. I should not have eaten that slice of pizza; I will probably have heartburn later tonight.
3. I should not have eaten that slice of pizza; it had mushrooms on it, and they always give me a headache.
4. I should not have eaten that slice of pizza. Who knows what kinds of preservatives it has? I'm going to get cancer.

Third Basic: Supporting Details in Cause and Effect

Once you have identified the causes or effects of an event or action, provide details that will help your reader understand these causes or effects. Details may include examples, descriptions, statistics, definitions, or expert opinions.

PRACTICE 12-6 Giving Examples and Details

Write down two causes or two effects for two of the three topics listed below. Then give an example or detail that explains each cause or effect.

Answers will vary.

Example

Topic: Bankruptcy

Cause 1: Overspending

Example/Detail: bought a leather jacket I liked and charged it

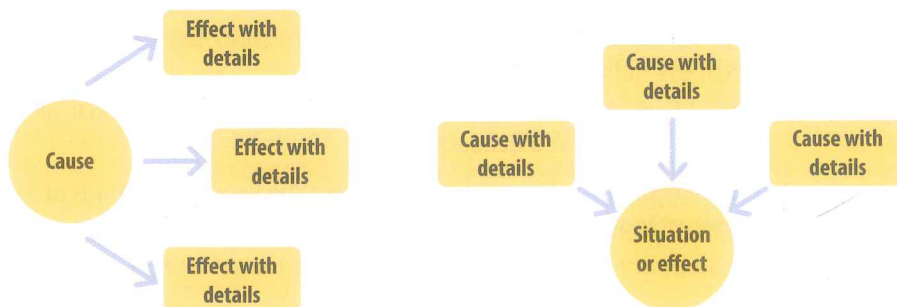
Cause 2: Poor budgeting

Example/Detail: not tracking monthly expenses versus income

1. A fire in someone's home
2. An A in this course
3. A car accident

Use a Graphic Organizer

Cause/effect graphic organizers help you to analyze major and minor support in a reading assignment or organize your prewriting for a writing assignment. One type of graphic organizer for cause/effect is a chain diagram, such as the one used by Jackie (p. 247). Here are two other common graphic organizers for cause and effect:



PRACTICE 12-7 Use Graphic Organizers in Reading

For each text, select and complete an appropriate graphic organizer:

1. The sample paragraph on pages 245–46.
2. The sample paragraph on page 256.
3. The essay by Tyler Dashner on pages 260–64.

PRACTICE 12-8 Use Graphic Organizers in Prewriting

Select one of the topics in Practice 12-6. Use a graphic organizer to illustrate your ideas, adding details if needed.

Fourth Basic: Organization in Cause and Effect

Cause and effect can be organized in a variety of ways, depending on the writer's purpose. When you are reading, use organization to help you clarify the writer's purpose. For example, if a writer wanted to show how a single event (loss of a cell phone) led to a chain of causes and effects that ended with being fired, a chronological order would be appropriate. However, if the writer's purpose is to encourage readers to avoid junk food, he would probably arrange his effects by order of importance. When you are writing, make sure your organization supports your purpose and your main idea.

Main Point	Purpose	Organization
The "Occupy" protests of 2011 brought attention to the economic difficulties faced by low- and middle-income citizens.	to explain the effects of the protests	order of importance, saving the most important effect for last
A desire to remain at a protest site for an extended period led "Occupy" protesters to create miniature towns, with food services, libraries, and more.	to describe the places where protesters camped out	space order
The "Occupy" protests in New York City inspired other protests throughout the country.	to describe the spread of the protest movement over time	time order

Transitions move readers smoothly from one cause to another, from one effect to another, or from causes to effects. Because cause and effect can use any method of organization depending on the writer's purpose, the following list shows just a few of the transitions you might see in cause/effect essays.

Common Transitions in Cause and Effect

	Conjunctive Adverbials	Subordinating Conjunctions	Adjective + Noun	Prepositions
Cause		because since	a primary cause a serious cause the most important cause one cause	because of due to
Effect	therefore as a result consequently		a long-term or short-term effect a primary effect a serious effect one effect the most important effect	

For more information on using and punctuating these transitions, see Chapters 17 and 24.

PRACTICE 12-9

Use Transitions in Cause and Effect

Read the paragraph that follows, and fill in the blanks with transitions. You are not limited to the ones in the preceding box. **Answers may vary. Possible answers are shown.**

Recently, neuroscientists, who have long been skeptical about meditation, confirmed that it has numerous positive outcomes. One, The first is that people who meditate can maintain their focus and attention longer than people who do not. This ability to stay “on task” was demonstrated among students who had been practicing meditation for several weeks. They reported more effective studying and learning because they were able to pay attention. Another, A second positive outcome is the ability to relax on command. Many people lead busy, stressful lives with multiple pressures on them—family responsibilities, work duties, financial worries, and uncertainties about the future. While meditating, people learned how to reduce their heart rates and blood pressure so that they could relax more easily in all kinds of situations. A third, The most important outcome is a thickening of the brain’s cortex. Meditators’ cortices were uniformly thicker than nonmeditators’. Because the cortex enables memorization and the production of new ideas, this last outcome is especially exciting, particularly in fighting Alzheimer’s disease and other dementias.

PRACTICE 12-10**Identify Causes and Effects and Create Sentences**

Identify the cause (C) and effect (E) in each of the following pairs of sentences. Then, combine the sentences into a single sentence using the transition word provided. Pay attention to punctuation, sentence structure, and pronouns as you combine sentences.

Example:

C Carlos and Bjorn come from different countries.

E Carlos and Bjorn speak English with different accents. (Because)

Combined sentence: Because Carlos and Bjorn come from different countries, they speak English with different accents.

1. C Ali arrived in the United States just four months ago, leaving his family behind and friends behind.
E Ali is facing culture shock and loneliness as he adjusts to his new country. (therefore)
2. C Ali's high school English teachers were British.
E Ali uses British spelling in his writing. (as a result)
3. E Ali struggles to write and submit his work online.
C Ali's high school did not use computers, tablets, or other technology. (because)
4. E Ali has beautiful handwriting.
C Ali has years of practice writing by hand using cursive script. (due to)
5. C Classroom instructors in the United States often face the board while talking.
E Ali doesn't always hear what his professors say.
C Classroom instructors in the United States speak rapidly. (consequently)
6. E Ali relies on an electronic dictionary and Google Translate.
C The vocabulary in the assigned readings is very difficult for Ali to understand. (since)
7. C Before coming to the United States, Ali had never seen an ice storm or a snow storm.
E He was afraid to drive after the first major snowstorm of the winter. (as a result)
8. C Over a foot of snow fell.
E Ali's college closed for two days. (because of)
9. E Ali could take his time completing the assignments.
C During the storm, Ali's assignments were posted online. (therefore)
10. C Ali felt relaxed while completing assignments at home during the snow storm.
E Ali has decided to take his next English class online. (Because)

Paragraphs vs. Essays in Cause and Effect

For more on the important features of cause and effect, see the Four Basics of Good Cause and Effect on page 245.

Paragraph Form

A Regular exercise improves cardiovascular health. One
B benefit of exercise is that it strengthens the heart. Like any
other muscle, the heart becomes stronger with use, and is able
to pump blood through the body more efficiently. The result
can be lower blood pressure, reducing the risk of heart disease.
C Another benefit of exercise is that it lessens the toll that ex-
cessive weight can take on the heart. In seriously overweight
individuals, the strain of carrying extra pounds can cause the
heart to enlarge, interfering with its ability to pump blood. By
losing weight through exercise and dietary changes, people can
reduce the burden on their hearts and also their cardiovascu-
D lar risk. The most important cardiovascular benefit of exercise
is that it lowers the risk of heart disease. As previously noted,
exercise can reduce blood pressure and strain on the heart, both
risk factors for heart attack, stroke, and heart failure. In addi-
tion, it can lower levels of “bad” cholesterol while raising levels
of “good” cholesterol. Controlling bad cholesterol is important
because when there is too much of this substance in the blood,
it can build up on artery walls, causing reduced blood flow.
E Regular and vigorous aerobic exercise is the best way to reap
these cardiovascular benefits, but even a brisk walk a few times
a week is better than no activity at all.

Main Point: Often narrower for a paragraph than for an essay: while the topic sentence (paragraph) focuses on just one major benefit of exercise, the thesis statement (essay) considers multiple benefits.

Support for the Main Point (Statements of Cause or Effect)

Paragraph organized by order of importance.

Detailed Explanations or Examples of Cause / Effect Statements: Usually 1 to 3 sentences per statement for paragraphs and 3 to 8 sentences per statement for essays.

- A** Topic sentence
- B** Support 1 (cause 1 or effect 1)
- C** Support 2 (cause 2 or effect 2)
- D** Support 3 (cause 3 or effect 3)
- E** Concluding sentence

Conclusion

Think Critically as You Read and Write Cause and Effect

Ask yourself as you read:

- Is the writer focused on causes, effects, or both?
- What is the writer's purpose? Is it clear?
- Are the causes and effects actual causes or effects? Are they logically connected?
- Does the writer provide relevant, well-balanced evidence for these causes and effects?

Ask yourself as you write:

- Have I examined a variety of possible causes and/or effects related to my topic? If not, research them, and consider revising your main point and support based on what you learn (see Chapter 14).
- Am I certain that my causes are real causes and my effects real effects? Are they logically connected?

Essay Form

Regular Exercise Has Multiple Benefits 1

Most people know how hard it is to start and stick with an exercise program. However, there is a good reason to build a significant amount of physical activity into every week: Regular exercise provides more physical and mental benefits than any medication could offer.

First, exercise helps people achieve and maintain a healthy weight. A nutritious diet that is not excessive in calories has a greater effect on weight loss than exercise does. However, regular exercise—ideally, interspersed throughout the day—can make an important contribution. For instance, people trying to lose weight might walk to work or to other destinations instead of driving. Or, they might take the stairs to their office instead of the elevator. If they go the gym at the end of the day, so much the better. Added up, all these efforts can make a difference.

Second, exercise boosts mood and energy levels. For example, exercise causes the body to release endorphins, chemicals that give us a sense of well-being, even happiness. Accordingly, exercise can help reduce

stress and combat depression. In addition, because exercise can make people look and feel more fit, it can improve their self-esteem. Finally, by improving strength and endurance, exercise gives individuals more energy to go about their lives.

The most important benefit of exercise is that it can help prevent disease. For example, exercise can improve the body's use of insulin and, as noted earlier, help people maintain a healthy weight. Therefore, it can help prevent or control diabetes. Additionally, exercise can lower the risk of heart attacks, strokes, and heart failure. For instance, exercise strengthens the heart muscle, helping it pump blood more efficiently and reducing high blood pressure, a heart disease risk factor. Also, exercise can lower levels of "bad" cholesterol while raising levels of "good" cholesterol. Controlling levels of bad cholesterol is important because when there is too much of this substance in the blood, it can build up in the walls of arteries, possibly blocking blood flow. Finally, some research suggests that

regular exercise can reduce the risk of certain cancers, including breast, colon, and lung cancer.

In my own life, exercise has made a huge difference. Before starting a regular exercise program, I was close to needing prescription medications to lower my blood pressure and cholesterol. Thanks to regular physical activity, however, both my blood pressure and cholesterol levels are now in the normal range, and I have never felt better. Every bit of time spent at the gym or exchanging a ride in an elevator for a walk up the stairs has been well worth it.

Essay organized by order of importance.

- A Thesis statement
- B Topic sentence 1 (first cause or effect)
- C Topic sentence 2 (second cause or effect)
- D Topic sentence 3 (third cause or effect)
- E Concluding paragraph

Read and Write Cause and Effect: Analyze and Respond

Student Cause and Effect

Caitlin Prokop

A Difficult Decision with a Positive Outcome



- Read the title and the first sentence. Will this essay address causes, effects, or both? How do you know?
- What are some difficult decisions you have made?



Guidelines for Reading and Annotating Cause and Effect Read and annotate this paragraph using these guidelines:

1. *Pay attention to vocabulary.* Use marginal notes, context clues, or a dictionary.
2. *Find the main point and underline it.* Remember that the main point shows the writer's purpose (to show causes or effects).
3. *Identify supporting details and organizational strategies.* Number the causes or effects, and determine whether these are organized chronologically or by order of importance.
4. *Ask questions and make connections.* Remember to connect the causes or effects to your own experiences and other things you have read.

* * *

When my mother made the decision to move back to New York, I made the choice to move in with my dad so that I could finish high school. This decision affected me in a positive way because I graduated with my friends, built a better relationship with my father, and had the chance to go to college without leaving home. Graduating with my friends was very important to me because I have known most of them since we were in kindergarten. It was a journey through childhood that we had shared, and I wanted to finish it with them.

Accomplishing the goal of graduating from high school with my close friends, those who accompanied me through school, made me a stronger and more confident person. Another good outcome of my difficult decision was the relationship I built with my dad. We never saw eye to eye when I lived with both of my parents. For example, we stopped talking for five months because I always sided against him with my mom. Living together for the past five years has made us closer, and I **cherish** that closeness we have developed. Every Thursday is our day, a day when we talk to each other about what is going on in our lives, so that we will never again have a distant relationship. A third good outcome of my decision is that I can go to Brevard Community College, which is right down the street. In high school, I had thought that I would want to go away to college, but then I realized that I would miss my home. By staying here, I have the opportunity to attend a wonderful college that is preparing me for transferring to a four-year college and finding a good career. I have done some research and believe I would like to become a police officer, a nurse, or a teacher. Through the school, I can do volunteer work in each of these areas. Right now, I am leaning toward becoming a teacher, based on my volunteer work in a kindergarten class. There, I can explore what grades I want to teach. In every way, I believe that my difficult decision was the right one, giving me many opportunities that I would not have had if I had moved to a new and unfamiliar place.

cherish: to value highly



Review your reading. Make sure you can answer these questions:

1. What are Prokop's main point and purpose in this paragraph?
2. How has she organized the paragraph? How do you know?
3. Does the paragraph illustrate the Four Basics of Good Cause and Effect? Explain.



1. Has Prokop provided enough details for a reader to understand the results of her decision? Explain.
2. Have you ever made a difficult decision that turned out to be a good one? Explain. When and how did you recognize the positive effects?

* * *

Tyler Dashner

A Look at Academic Dishonesty

Preview

Read

Review

Reflect and Respond

- Read the title and skim the first paragraph. What do you think this paper will address? Do you think the writer is concerned about causes, effects, or both? How do you know?
- How would you define “academic dishonesty”? Do you think your instructors and friends work from the same definition?
- How common do you think academic dishonesty is?

Preview

Read

Review

Reflect and Respond

Read and annotate this essay using the Guidelines for Reading and Annotating Cause and Effect on page 258.

I was thinking up topics for this paper when I stepped into one of my other classes. Another student asked me for an answer on a take-home quiz we had been assigned. I politely said that it was a quiz, and I was not comfortable sharing my answers. Everyone in the class gave me a funny look, as if I was the bad guy for not helping this other student. In my mind, breaking the “Academic Honesty Pledge” was not something I was willing to do. Judging from the response of almost every other student in the room though, I began to wonder how the average student felt about this topic.

2 I began asking myself some basic questions on academic honesty. I had never given it a large amount of thought. Would I cheat if given the opportunity? Would I not cheat due to morals and ethics? To be totally honest I came to a very arrogant conclusion: I don’t cheat because I know I can do better work than the students around me; cheating would lower my grade, not raise it. When it comes to critical writing, there is a wealth of information available from writers who are much more experienced and more **articulate** than I. What keeps me from plagiarizing these expert writers?

3 “Academic dishonesty involves, but is not limited to, one of the following: giving or receiving, offering or soliciting unauthorized assistance on any exam or assignment; plagiarism or **collusion**” (LFCC Student Handbook). This is Lord Fairfax Community College’s official description of academic dishonesty and the definition I will be using in my paper when referring to cheating or academic dishonesty.

articulate: to express something in a way that can be clearly heard and understood

collusion: secret cooperation for an illegal or dishonest purpose

4 While doing research on this topic, I noticed an obvious pattern. A large majority of academics conducting research on this topic felt that teachers create an environment that either encourages or discourages cheating. If a teacher is honest with his or her students (e.g., sticking to the syllabus and grading fairly), this promotes a trusting environment and has been shown to discourage cheating (Hulsart and McCarthy 93). I have noticed this phenomenon for myself. When I have professors who are engaging and positive and have a clear syllabus that they stick to, I find that doing the work for the class comes easily. When the professor is constantly behind on the syllabus and puts down students, I always have trouble staying motivated to put forth one hundred percent effort in the class.

5 It isn't just the atmosphere in the classroom that propagates the mentality to cheat; "[s]ocial norms theory says that people tend to maintain behavior consistent with peer **descriptive norms**, and that overestimating the frequency that one's peers engage in a behavior can lead to increases in that behavior" (Hard, Conway, and Moran 1059). This theory involves the idea that perception is reality. If students think everyone is cheating, then they will be more inclined to cheat themselves in order to keep up academically with their peers. This makes it incredibly difficult for educators to address cheating. If they discourage it, students will think everyone is doing it, so they should too. If they do nothing about it, students will feel they can cheat without repercussions.

6 This perception of cheating was exemplified by a Scanlon and Neumann study that found that eight percent of students admitted to cutting and pasting directly from sources, whereas fifty percent of students felt that other students were doing this behavior (380). Clearly students believe that cheating is more prevalent than it really is. This cycle, if left unchecked, should be spiraling out of control, but it's clearly not. Researchers say that the Internet is making it easier to cheat, but the number of students cheating remained fairly constant from the 1980s to the 1990s (Scanlon and Neumann 381–82). Still, students with a **propensity** to cheat will cheat by any means available to them (Witherspoon, Maldonado, and Lacey 82).

7 Zwagerman asks his fellow academics to address plagiarism differently. He feels that professors are concerned only with catching cheaters and punishing them. Plagiarism is a result of students who see their futures dependent on their grades and, out of fear, cheat to ensure they get good grades. Zwagerman says that a poor student will cheat poorly and get caught, but a good student will cheat better and most likely not get caught. In the end, the grade is the same. Academia's **call to arms** against plagiarism hurts students and faculty by destroying any trust between them. Teachers should educate their students about plagiarism instead of threatening them (Zwagerman).

8 After thinking all of this over and doing research on the subject, I began to realize just how complex the issue truly is. I came up with a series of questions that I wanted to see how students at LFCC would answer. I surveyed thirty

descriptive norms: people's perceptions of what is commonly done in specific situations

propensity: a strong natural tendency to do something

call to arms: a summons, invitation, or appeal to undertake a particular course of action

students randomly in the cafeteria of the school and some in various classes of mine. I asked them a series of questions:

1. On a scale of 1 to 10, 1 being solely for a degree and 10 being solely for an education, how would you rate your reason for attending Lord Fairfax Community College?
2. On a scale of 1 to 10, 1 being my professors grade fairly and 10 being my professors have no rhyme or reason to their grading, how would you rate your professors' grading standards at Lord Fairfax Community College?
3. On a scale of 1 to 10, 1 being my professors always follow their syllabus and 10 being my professors never follow their syllabus, how would you rate your professors on how well they follow their syllabus at Lord Fairfax Community College?
4. On a scale of 1 to 10, 1 being my professors set a clear standard for assignments and 10 being my professors are very ambiguous about what they want from their students, how would you rate your professors' expectations on assignments at Lord Fairfax Community College?
5. Knowing that you would not get caught and that you would receive a better grade on an assignment or a test, would you cheat in some way (e.g., plagiarize or copy off another student)?

9 I then spoke with them afterward to clarify their results and to gain further insight into how these issues affected them. Of the students I surveyed, they were all surprisingly upfront about these issues. This was due to the fact that I assured each student that no professor would ever see their surveys and I would destroy them after completing this assignment. Without these assurances, I feared that students would be apprehensive about giving me the candid responses I would need to legitimize my data.

10 I would have liked to have been able to ask more in-depth questions, but I felt that a shorter survey would get a better sampling of the average college student, instead of just getting the more motivated students (i.e., the "good" students), thus skewing the results. Students in the cafeteria did not always have the time to fill out a lengthy survey, especially those students that had **procrastinated** and were desperately trying to finish an assignment for an upcoming class (i.e., the "bad" students).

11 I added up the numbers the students provided for each question and averaged them. I used this to give me an idea of how the average student felt about these topics. I also **tabulated** how the response to each of the first four questions affected the answer to the final question. I feel that both sets of data were necessary to determine the mind-set of a student when confronted with the option of cheating.

12 The first question's average was 2. I felt this question was important because I wanted to know what effect a student's reason for attending college had on his or her propensity to cheat. I also was curious to see if students at LFCC were interested in an education at all. From talking to students, I came to the conclusion that an education was a means to an end. Students need a degree in order to get

procrastinated: to be slow or late about doing something that should be done

tabulated: to arrange information in an organized way so that it can be studied, recorded, etc.

the job they want, and in most career fields a large majority of the information they need to know is gained from on-the-job training.

13 The second question's average was 5. From interactions I had with the participants after the survey I have determined this was because students felt that about half of their professors graded fairly and half did not grade fairly. Also from their feedback I determined this to be the most discouraging thing for students when it came to their college experience.

14 The third question's average was 3. Students felt that their professors almost always followed their syllabuses. If professors needed to make adjustments they would either amend the syllabus, making the new syllabus available on Blackboard, or they would clarify the changes in class. Students did mention that this would be frustrating if it happened to them. Judging from these results I determined that the instructors at LFCC do a good job of issuing syllabuses and keeping to the schedule they lay out at the beginning of the semester.

15 The fourth question's average was 6. Most students responded to this question with the complaint that teachers at LFCC purposely make the assignments and tests at the beginning of the semester hard or ambiguous in order to "weed out" the students who won't be taking the class seriously and who will eventually drop it. This could be frustrating to these students, but the professors usually had extra credit and/or an abundance of points throughout the semester to compensate for this initial bad grade on an assignment or test. The feelings on this subject were split pretty evenly; some students felt this was unfair, while others understood the rationale and adapted to their professors' individual teaching styles.

16 The fifth and final question I left as a simple yes/no question. Only seven people answered yes and twenty-seven answered no. Interestingly, the students that answered yes to this question were also more critical of their professors in the previous questions, while the students that answered no were either equally as critical or far less critical of their professors. This can be explained because for some students, an ethical code keeps them from cheating, regardless of the professor's behavior. I also believe that there is some margin for error on the final question due to students' apprehension about admitting to a random person that they are willing to cheat.

17 Judging from these results, my data confirms what other researchers have been saying. The way a professor conducts a class has an impact on a student's propensity to cheat. Although there is no way to keep every student from cheating, by exercising the same respect toward students as is expected from them in return, professors can **mitigate** the amount of dishonest behavior among their students. Professors can also inform their students that cheating will not be tolerated and ensure that all students have read the academic dishonesty guidelines.

18 From my research it is obvious that administrators at LFCC understand this research and have implemented most of the recommendations made by the research I read. I now understand why on the first day of class every one of my professors has the same routine of going over the syllabus and ensuring each student understands all of the school policies toward academic dishonesty. These simple steps clearly mitigate the amount of students who cheat without degrading students by treating them all like cheaters.

mitigate: to make something less severe, harmful, or painful

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Review your reading. Make sure you can answer the following questions:

1. What is Dashner's main point?
2. What are the causes of academic dishonesty, according to the paper?
3. What evidence does Dashner provide for the causes he mentions?
4. Does the paper demonstrate the Four Basics of Good Cause and Effect? Explain.

Now, using your annotations, write a summary of Dashner's essay, following the guidelines from Chapter 6.

Preview

Read

Review

Reflect and Respond

1. In this paper, the writer begins by referring to scholarly sources, but he also cites the results of an informal survey he conducted on his campus. How do the questions he asked in his survey relate to the scholarly research he mentions? Would the paper have been as effective if he had left out the informal survey? Explain.
2. Does he provide enough evidence for the causes he presents? Explain.
3. Do you think he left out any possible causes of academic dishonesty? Explain.
4. Are there situations in which cheating might be justified? Explain.
5. Craft a survey similar to Dashner's to investigate attitudes and habits of cheating on your campus. Report your results in a paragraph or an essay that describes either the causes or the effects of cheating.

Professional Cause and Effect

John Tierney

Yes, Money Can Buy Happiness

Preview

1. Skim the title, the information about the author, and the first paragraph. What is the rhetorical context (audience, purpose, and topic) for this essay?
2. Does this essay address causes, effects, or both? How do you know?
3. Create a guiding question for your reading, based on your previewing.

Read and annotate this essay carefully, following the Guidelines for Reading and Annotating Cause and Effect on page 258.



FRED R. CONRAD/THE NEW TIMES/REDUX

A well-known columnist for the *New York Times* since 1990, John Tierney has an extensive background in news writing. After graduating with a degree in American studies from Yale University, Tierney reported for a series of publications, including the *Bergen Record*, the *Washington Star*, and *Science* magazine. He then worked for several years as a freelance writer, reporting on six continents and publishing articles in more than fifteen national newspapers and magazines. His book, *Willpower: Rediscovering the Greatest Human Strength*, written with Roy Baumeister, was named one of Amazon's Best Books of 2011.

correlated: is related or connected to something else

Yes, money can buy happiness, but probably not in the way you imagined. Spending it on yourself may not do much for your spirits, but spending it on others will make you happier, according to a report from a team of social psychologists in the new issue of *Science*.

2 The researchers confirmed the joys of giving in three separate ways. First, by surveying a national sample of more than 600 Americans, they found that spending more on gifts and charity **correlated** with greater happiness, whereas spending more money on oneself did not. Second, by tracking sixteen workers before and after they received profit-sharing bonuses, the researchers found that the workers who gave more of the money to others ended up happier than the ones who spent more of it on themselves. In fact, how the bonus was spent was a better predictor of happiness than the size of the bonus.

3 The final bit of evidence came from an experiment in which forty-six students were given either \$5 or \$20 to spend by the end of the day. The ones who were instructed to spend the money on others—they bought toys for siblings, treated friends to meals, and made donations to the homeless—were happier at the end of the day than the ones who were instructed to spend the money on themselves.

4 “These experimental results,” the researchers conclude, “provide direct support for our causal argument that spending money on others promotes happiness more than spending money on oneself.” The social psychologists—Elizabeth Dunn and Lara Aknin of the University of British Columbia, Vancouver, and Michael Norton of Harvard Business School—also conclude that “how people choose to spend their money is at least as important as how much money they make.”

5 I asked Dr. Dunn if she had any advice for readers on how much to spend on others. Her reply was, “I think even minor changes in spending habits can make a difference. In our experiment with college students, we found that spending just \$5 prosocially had a substantial effect on happiness at the end of the day. But I wouldn’t say that there’s some fixed amount that everyone should spend on others. Rather, the best bet might be for people to think about whether they can push themselves to devote just a little more of their money to helping others.”

6 But why wouldn’t people be doing that already? Because most people don’t realize the personal benefits of charity, according to Dr. Dunn and her colleagues. When the researchers surveyed another group of students, they found that most of the respondents predicted that personal spending would make them happier than spending the money on other people.

7 Perhaps that will change as word of these experiments circulates—although that prospect raises another question, which I put to Dr. Dunn: If people started giving away money chiefly in the hope of making themselves happier, as opposed to wanting to help others, would they still derive the same happiness from it?

8 “This is a fascinating question,” she replied. “I certainly hope that telling people about the emotional benefits of prosocial spending doesn’t completely

erase these benefits; I would hate to be responsible for the downfall of joyful prosocial behavior.”

9 Do you have any theories on the joys of giving? Any reports of your own experiments? Or any questions you’d like to ask the researchers? Dr. Dunn, in keeping with the results of her experiments, has generously offered to provide some answers free of charge.

Check Comprehension

1. What is Tierney’s main point?
2. What evidence does he provide for his point?
3. What question does Tierney raise about the main point at the end of his essay?

Think Critically

Use clues and information from the reading to answer the following questions:

1. Who is Tierney’s audience? What does he expect his audience to believe about money and happiness?
2. Does Tierney believe he has answered all of the readers’ questions about money and happiness?
3. What would Tierney recommend that his readers do to improve their happiness?

Summarize

Use your notes and annotations to summarize this essay, following the guidelines in Chapter 6.

Writing Strategies in Action: Vary Use of Quotes

In his essay, Tierney quotes experts, a strategy you have seen several times in other readings. You have already learned how to introduce a quote with a signal phrase (see p. 133). Tierney’s essay illustrates a variation on this basic pattern.

Variation 1: Move the Signal Phrase

In paragraph 4, Tierney quotes three researchers. Notice where he puts the signal phrase. Then think about the questions that follow (for more information on quotation marks, punctuation, and capital letters, see Chapters 30–32).

“These experimental results,” **the researchers conclude**, “provide direct support for our causal argument that spending money on others promotes happiness more than spending money on oneself.”

1. What happens to capital letters when the signal phrase interrupts a complete quote?
2. How are commas used when the signal phrase interrupts a complete quote?
3. What happens to the flow or rhythm of a sentence when a signal phrase interrupts a complete quote? (Read it out loud.)
4. Could the signal phrase occur in a different place in the sentence? Why or why not?

Rewrite to Move the Signal Phrase

Rewrite each of the following quotes, moving the signal phrase to a different part of the sentence each time. Check your punctuation carefully, and read your sentence aloud. Which version is better, the original or the revision?

1. According to Tierney, "The final bit of evidence came from an experiment in which forty-six students were given either \$5 or \$20 to spend by the end of the day" (266).
2. Tierney asks, "But why wouldn't people be doing that already?" (266).
3. John Tierney claims, "Yes, money can buy happiness, but probably not in the way you imagined" (266).

Explore Grammar: Use By-Modifiers Correctly

Look at the following sentences from Tierney's essay:

First, **by surveying a national sample of more than six hundred Americans,** they found that spending more on gifts and charity correlated with greater happiness...

Second, **by tracking workers before and after they received profit-sharing bonuses,** the researchers found that the workers who gave more of the money to others ended up happier...

The bold part of each sentence is a modifier; it explains how the subject of the sentence completed the action of the sentence. Notice each sentence has a subject (underlined once) and a verb (underlined twice). The modifier helps clarify the meaning of the sentence, but it does not replace the subject or the verb. When you begin a sentence with a "by" modifier, be sure not to omit the subject of the sentence. Be sure to put a comma after the modifier.

- Incorrect** By studying the notes several days before the test can improve your score.
- Correct** By studying the notes several days before the test, **a student** can improve his or her score.
- Incorrect** By giving money to charitable organizations can save on taxes.
- Correct** By giving money to charitable organizations, **taxpayers** can save on taxes they owe.

Think, Discuss, and Respond

1. The scientist in the article says she hopes that people will “think about whether they can push themselves to devote just a little more of their money to helping others.” Will this article change your attitude about charity or cause you to alter your behavior in any way? Write a paragraph explaining your answer.
2. Tierney’s article explores “the emotional benefits of prosocial spending.” Have you ever experienced these benefits? Write an essay to explore an experience in which you spent money on or helped others. Did the action cause the effects that scientists found in their studies?

* * *

Steven Pinker

Are Your Genes to Blame?

Preview

- Skim the title and the first line of each paragraph. What can you learn about the rhetorical context (audience, purpose, and topic) for this essay?
- Look at paragraph 9. Is this essay about causes, effects, or both? How do you know?
- Create a guiding question for your reading of this essay.

Read and annotate this essay carefully, following the Guidelines for Reading and Annotating Cause and Effect on page 258.



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Steven Pinker (b. 1954) is an experimental psychologist, language researcher, and cognitive scientist who writes about science and language for publications such as the *New York Times*, *TIME*, and *The New Republic*. He has written several books, including *How the Mind Works* (1997) and *The Better Angels of Our Nature* (2011), and is considered one of the most influential thinkers of our time. Pinker teaches psychology at Harvard University.

introversion: a personality trait often characterized by shyness and a preference for not being in the company of others

conscientiousness: attention to what is right and wrong; following one's conscience in a careful and consistent way

antagonism: a clear and obvious hostility between people or groups

susceptibility: an ability to be affected by something, such as a disease

Brave New World: title of a book published in 1932 by Aldous Huxley that describes a fictional world in which reproduction, social order, and economics are managed and tightly controlled by the state

genome: a full set of human chromosomes containing all genetic material passed from one generation to the next

erode: reduce or wear away

Study after study has shown that genes can affect behavior and mental life. Identical twins separated at birth (who share their genes but not their environment) are similar in their intellectual talents, their personality traits (such as **introversion**, **conscientiousness**, and **antagonism**), their average level of life-long happiness, and personal quirks such as giggling incessantly or flushing the toilet both before and after using it. Identical twins (who share all their genes) are more similar than fraternal twins (who share half their genes). And biological siblings (who share half their genes) are more similar than adopted siblings (who share none of their genes). It's not only personality and intelligence that are partly heritable, but **susceptibility** to psychological maladies such as schizophrenia, obsessive-compulsive disorder, and major depression.

2 The discovery that genes have something to do with behavior came as a shock in an era in which people thought that the mind of a newborn was a blank slate and that anyone could do anything if only they strove hard enough. And it continues to set off alarm bells. Many people worry about a **Brave New World** in which parents or governments will try to re-engineer human nature. Others see genes as a threat to free will and personal responsibility, citing headlines such as "Man's genes made him kill, his lawyers claim." Behavioral geneticists are sometimes picketed, censored, or compared to Nazis.

3 With increasing knowledge of how the **genome** works, many beliefs about ourselves will indeed have to be rethought. But the worst fears of the genophobes are misplaced. It is easy to exaggerate the significance of behavioral genetics for our lives.

4 For one thing, genes cannot pull the strings of behavior directly. Behavior is caused by the activity of the brain, and the most the genes can do is affect its wiring, its size and shape, and its sensitivity to hormones and other molecules. Among the brain circuits laid down by the genes are ones that reflect on memories, current circumstance, and the anticipated consequences of various courses of action and that select behavior accordingly, in an intricate and not entirely predictable way. These circuits are what we call "free will," and providing them with information about the likely consequences of behavioral options is what we call "holding people responsible." All normal people have this circuitry, and that is why the existence of genes with effects on behavior should not be allowed to **erode** responsibility in the legal system or in everyday life.

5 Don't count on any single gene with a large behavioral effect being identified any time soon. Behavioral genetics has uncovered a **paradox**. Studies that measure similarities among twins and adoptees reliably show strong effects of sharing many genes (such as half a genome, or all of one). The outcome is so reliable that behavioral geneticists now speak of the First Law of their field: that all behavioral traits are partly heritable.

6 But studies that try to isolate a single gene for a behavioral trait have been **fickle**; many of the **putative** genes-for-X have not held up in replications. Genes must exert their effects by acting together in complex combinations. A rough **analogy**: a computer program can have a trait, such as being easy to use, without necessarily having a single magical programming instruction that makes any program easy to use when added and any program hard to use when omitted.

7 So psychological engineering is more remote than the futurologists would have you believe. Though musical talent may be partly heritable, there is probably no single gene for musical talent that ambitious parents can have implanted into their unborn children. It might take hundreds or thousands of the right genes, with a different combination needed for each child.

8 Finally, the fact that genes matter doesn't mean that other things don't matter. Some of these causes are obvious. There are no genes for speaking English or for being a Presbyterian (though there may be sets of genes for verbal skill and religiosity). These depend entirely on one's culture. Others are less obvious, such as germs, accidents, chance encounters in life, and random events in the development of the brain **in utero**.

9 And still other environmental factors may not be acting as we think they do. It's easy to assume that the variation in behavior that is not caused by genes must be caused by parents. But it's been surprisingly hard to demonstrate any long-term effects of growing up in a particular family within a culture. Identical twins reared together are similar, but they are not literally identical: one may be more anxious than the other, one may be gay and the other straight. This shows that genes are not everything—but since these twins grow up in the same family, it also shows that what isn't explained by genes isn't explained by family influences either. Similarly, children need to hear English to acquire it. But if their parents are immigrants, they end up with the accent of their peers, not their parents.

10 Though the effects of genes may be easy to overestimate, they are also easy to underestimate. Many failed **utopias** of the twentieth century dreamed of nurturing a "new man" free of selfishness, family ties, and individual differences. Some psychotherapists promise what they cannot deliver, such as transforming a shy person into a bold one or a sad sack into a barrel of monkeys.

11 None of this means that social and personal improvement are a waste of time. Even if each of us is born with a range of temperament and talent, we can try to reach the best point in that range. And even if we have a nature, part of that nature is an open-ended **ingenuity** that can think up possible solutions to our problems. Using our genes as an excuse for **fatalism** is unwise. But so is pretending that they don't matter at all.

paradox: a statement that contradicts itself; a situation in which two opposites are true at the same time

fickle: not consistent, likely to change

putative: supposed

analogy: a comparison used to illustrate a concept or situation

in utero: in the uterus; before birth

utopia: a perfect society or an attempt to create such a society

ingenuity: cleverness

fatalism: the belief that a person cannot change his or her nature or circumstances because everything is determined by fate

Check Comprehension

1. What is the main idea of this essay?
2. Why are people afraid of research discussing behavior and genetics? (para. 2)
3. What are three reasons we should not be afraid of researching genetics and behavior? (paras. 4, 5, and 8)
4. What other factors influence behavior? (para. 9)
5. What is a rational response to the influence of genetics on behavior and ability, according to Pinker? (para. 11)

Think Critically

Based on your understanding of the essay, what would Pinker say to the following students?

1. Bethany: "I am destined to fail math; after all, my parents weren't good math students, and I inherited their lousy math skills."
2. Darrin: "I would love to play professional football, but I'm 5'4" and I weigh only 160 pounds. I guess football isn't a career option that is open to me."

Writing Strategies in Action: Vary Use of Quotes

You have learned how to quote a complete sentence and how to vary the structure of a quote by moving the signal phrase (see p. 267). Another way to use a quotation is to quote only *part* of a sentence.

Variation 2: Quote only Part of the Original Sentence

Let's look at an example. One student responded to a question about this article by writing this sentence:

Pinker explains that the genes we inherit from our parents **"cannot pull the strings of behavior directly"** (270).

Notice how this sentence is different from other examples you have seen:

1. Only part of the original sentence is quoted (and the quoted part is not a complete sentence).
2. The quoted part is not introduced by a signal phrase. Instead, it fits into the structure of the whole sentence.
3. The first letter of the first word in the quote is not capitalized.

Quoting only part of a sentence—or a phrase—is another way to use quoted material in your own writing. Take a look at one more example:

Steven Pinker describes two fears concerning behavioral genetics: society's ability to **"reengineer human nature"** and the potential of using genes as an excuse for crimes and other forms of bad behavior (270).

Here, only three words are copied from the original essay; the rest of the sentence is actually a paraphrase. But since the ideas and some specific words are borrowed from Pinker's essay, we still include a citation.

Use Partial Quotations

Answer each of the following questions. In each answer, be sure to quote part of a sentence from Pinker's essay. Cite your quote properly, and use the examples as a model.

Example: What do studies of twins separated at birth tell us?

Answer: Pinker explains that identical twins who are not raised in the same home show how genes can affect intelligence, personality, happiness, and even "personal quirks, such as giggling incessantly or flushing the toilet both before and after using it" (270).

1. How do genes affect behavior? (para. 4)
2. How does Pinker view "free will"? (para. 4)
3. Can science give us a single gene connected to a single behavior? (para. 6)
4. What sorts of abilities or behaviors are determined by culture instead of genes? (para. 7)
5. Are family environment and genetics the only factors that influence behavior? (para. 8)
6. What is a reasonable attitude towards our genetic inheritance? (para. 10)

Paraphrase

In the exercise above, you answered a series of questions using partial quotes. Now answer the same questions, using paraphrases. Be sure to use a signal phrase and cite each paraphrase accurately.

Build Vocabulary: Prefixes

Many English words include a **prefix**, one or more letters added to the beginning of a word that change its meaning. Recognizing common prefixes, such as *pre-*, can help you understand new vocabulary.

Look at the following words that begin with the prefix *pre-*: What do you think *pre-* means?

prefix, preview, prepay, precaution, prehistoric, predict

Identify the prefix in each of the following words from Pinker's article. What does each prefix mean?

Word	Prefix	Meaning of Prefix
introversion (related words are introvert, introverted, and introspection)	intro-	Answers will vary. within, inwardly
reengineer, rethink, restate	re-	again
misplaced, mistook, misheard	mis-	wrong, mistaken, negative of root word
uncovered, unborn, unwise	un-	not
transforming, translated, trans-Pacific	trans-	across, through, beyond
implanted, impatient, impossible	im-	not, opposite of

Build Vocabulary: Affect and Effect

Steven Pinker uses both nouns and verbs to describe the effects of our genes on our behavior. Look at the following sentence from paragraph 4:

Behavior is caused by the activity of the brain, and the most genes can do is affect its wiring, its size and shape, and its sensitivity to hormones and other molecules.

Is the underlined word a noun or a verb? How do you know? Now look at this sentence, from paragraph 6:

Genes must exert their effects by acting together in complex combinations.

Is the underlined word a noun or a verb? How do you know?

Based on these sentences, how would you explain the difference between *affect* and *effect*? Can you find additional examples in Pinker's essay to support your answer?

Think, Discuss, and Respond

1. Pinker suggests that some of our behavioral and personality traits are inherited genetically, but others may not be. What is one trait that you believe you inherited from your parents genetically? What is one trait that you do not believe you inherited from your parents? Describe these two traits and explain why you believe you did (or did not) inherit each trait genetically.
2. In paragraph 5, the writer suggests that behavioral genetics presents us with a paradox. What is a paradox? What is the paradox that Pinker describes in paragraphs 5 and 6? Explain this paradox in your own words, making sure that you cite Pinker's ideas accurately.
3. Pinker provides an analogy in paragraph 6: genetic traits may be like a computer program. Write a paragraph explaining this analogy in your own words, making sure to cite your paraphrase accurately. Can you use another analogy to illustrate the same point?
4. In paragraph 4, Pinker defines both "free will" and "holding people responsible" in terms of circuits within our brains. How would you define these concepts? Explain your answer in a paragraph.

Extend and Connect: Write Your Own Cause-and-Effect Essay

1. In paragraph 2, Pinker describes society's fears about the science of behavioral genetics. What might cause someone to be afraid of science and scientific research? Explore your answer in an essay.
2. While genetic inheritance does impose some limits on a person's abilities, according to Pinker, that should not stop people from trying to improve themselves and make changes for the better. What can cause a person to make a significant change in behavior? What can prevent a person from making such a change? Explore these questions in an essay, referring to Pinker and at least one other essay in the text for support.
3. Tierney's essay explains how giving money to others can make us happy. What else can cause happiness? Explore the causes of happiness, using Tierney's essay and at least one other essay in the book.

CHECKLIST**Writing a Cause-and-Effect Essay****NARROW AND EXPLORE YOUR TOPIC (see Chapter 2)**

- ☐ Make the topic more specific.
- ☐ Use prewriting techniques to get ideas about your narrow topic.

**DRAFT A TOPIC SENTENCE OR THESIS STATEMENT (see Chapter 4)**

- ☐ Determine your purpose: do you want to analyze causes, effects, or both?

SUPPORT YOUR POINT (see Chapter 4)

- ☐ Prewrite to find causes (or effects), and make sure that each one is a real cause (or effect).
- ☐ Choose the most significant causes or effects.
- ☐ Discover details, definitions, and examples to support the causes or effects.

WRITE A FULL DRAFT (see Chapter 4)

- ☐ Make a plan that arranges the causes and effects logically.
- ☐ Include the topic sentence (paragraph), an introduction, a thesis (essay), and all the support points.
- ☐ Add a concluding sentence or concluding paragraph.

**REVISE YOUR DRAFT (see Chapter 5)**

- ☐ Make sure you have covered the Four Basics of Good Cause and Effect.
- ☐ Get feedback from your peers.
- ☐ Make sure all your causes and/or effects are directly related to the topic.
- ☐ Make sure you have included transitions to move readers smoothly from one cause/effect to the next.

**EDIT YOUR REVISED DRAFT (see Chapter 5)**

- ☐ Correct mistakes in grammar, spelling, punctuation, and word usage.
- ☐ Make sure your draft is formatted correctly.